Tutorial 1. Exercise GC1. Mixture, Matter, and Measurement

Exercise 1. ***(Kno)***

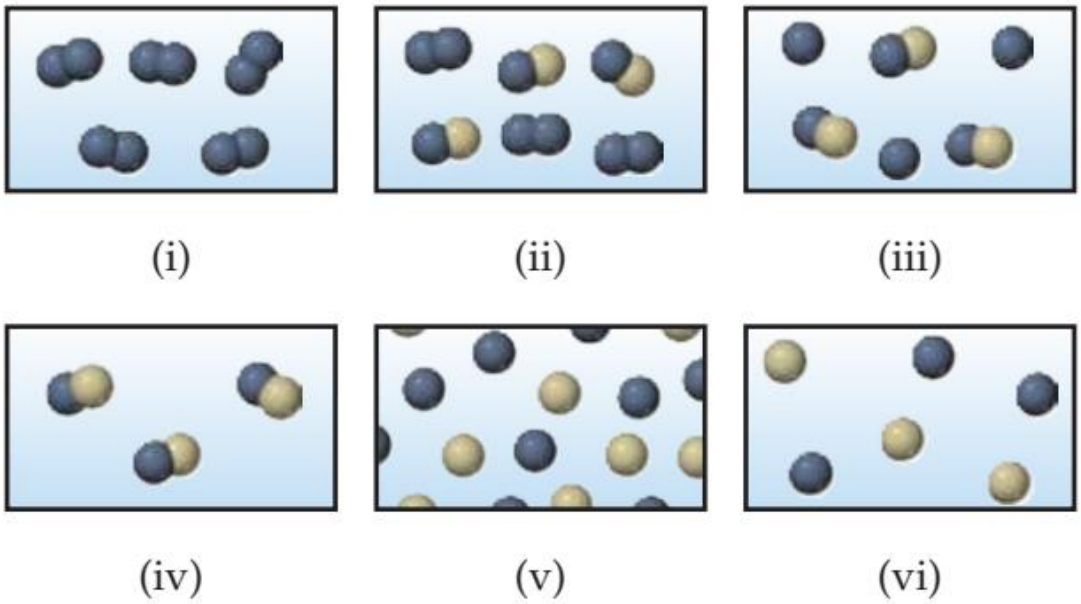
Which of the following figures represents? (More than one picture might fit each description.)

(**a**) a pure element,

(**b**) a mixture of two elements,

(**c**) a pure compound,

(**d**) a mixture of an element and a compound?



Exercise 2. ***(Kno)***

Classify each of the following as a pure substance or a mixture. If a mixture, indicate whether it is homogeneous or heterogeneous:

(**a**) milk,

(**b**) beer,

(**c**) diamond,

(**d**) mayonnaise

Exercise 3. ***(Und)***

A match is lit to light a candle. The following observations are made:

(a) The candle burns.
(b) Some wax melts.
(c) Melted wax solidifies on the candleholder.
(d) Soot (carbon) is produced by the burning of the match and the candle.

Which of these occurrences are due to physical changes, and which are due to chemical changes?

Exercise 4. ***(Und)***

Are the separation methods below best described as distillation, filtration, or chromatography?

a) Brewing of coffee:



b) Separating solvent from solution using rotary evaporator



Exercise 5. ***(Und)***

(**a**) Three spheres of equal size are composed of aluminum (density = 2.70 g/cm^3), silver (density = 10.49 g/cm^3), and nickel (density = 8.90 g/cm^3). List the spheres from lightest to heaviest.

(**b**) Three cubes of equal mass are composed of gold (density = 19.32 g/cm^3), platinum (density = 21.45 g/cm^3), and lead (density = 11.35 g/cm^3). List the cubes from smallest to largest.

Exercise 6. ***(Und)***

Identify each of the following as measurements of length, area, volume, mass, density, time, or temperature:

a) 25 ps,

(**b**) 374.2 mg,

(**c**) 77 K,

(**d**) 100,000 km^2,

(**e**) 1.06 mm,

(**f**) 16 nm^2,

(**g**) -78 °C,

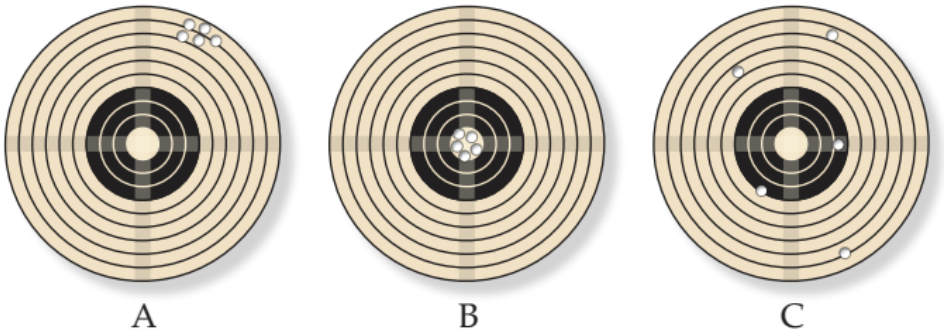
(**h**) 2.56 g/cm^3,

(i) 28 cm^3

(k) 100 ml

Exercise 7 ***(Und)***

The three targets from a rifle range shown here were produced by: (A) the instructor firing a newly acquired target rifle; (B) the instructor firing his personal target rifle; (C) a student who has fired his target rifle only a few times.



(a) Comment on the accuracy and precision for each of these three sets of results.

(b) For the A and C results in the future to look like those in B, what needs to happen?

Exercise 8 ***(App)***

(**a**) What is the mass of a silver cube whose edges measure 2.00 cm each at 25 °C? The density of silver is 10.49 g/cm^3 at 25 °C.

(**b**) The density of aluminum is 2.70 g/cm^3 at 25 °C. What is the weight of the aluminum foil with an area of 0.5 m^2 and a thickness of 0.5 mm?

(**c**) The density of hexane is 0.655 g/mL at 25 °C. Calculate the mass of 1.5 L of hexane at this temperature.

Exercise 9 ***(App)***

Make the following conversions:

(**a**) 83 °F to °C

(**b**) 29 °C to °F

(**c**) 294 °C to K

(**d**) 832 K to °C

Exercise 10 ***(Kno)***

Use appropriate metric prefixes to write the following measurements without use of exponents:

(**a**) 7.29×10^{6} g

(**b**) 6.1×10^{-10} m

(**c**) 1.828×10^{-3} s

(**d**) 3.523×10^{9} m^3

(**e**) 9.62×10^{2} m/s

(**f**) 8.923×10^{-12} kg

(**g**) 3.552×10^{12} L